

Connected particles

Strings, pulleys and tension

YOUR AIM TODAY

Build one force equation for each particle and solve simultaneously.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

Particles joined by a light inextensible string share the same acceleration magnitude.

- Light string means its mass is ignored.
- Smooth pulley means the tension is the same on both sides.
- Choose one direction of motion for the whole system.
- Write $F=ma$ separately for each particle.
- Add equations to eliminate tension; substitute back to find it.

MEMORY RULE

Same a , same T , separate equations.

Hanging pair

WORKED STEPS

Masses 3 and 2 kg: $3g - T = 3a$; $T - 2g = 2a$. Add: $g = 5a$, so $a = g/5$; then $T = 2g + 2a$.

Check

WORKED STEPS

T lies between the two weights for a simple hanging pair.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Masses 5 kg and 3 kg hang. Write both equations.	
2. Find acceleration.	
3. Find tension.	
4. A 4 kg table block connects to 2 kg hanging, smooth surfaces. Write equations.	
5. Find acceleration and tension for Q4.	
6. State two assumptions that make tension equal throughout.	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$5g - T = 5a$; $T - 3g = 3a$.
2	$2g = 8a$, so $a = g/4 = 2.45 \text{ m/s}^2$.
3	$T = 3g + 3a = 36.75 \text{ N}$.
4	For 4 kg: $T = 4a$. For 2 kg: $2g - T = 2a$.
5	$2g = 6a$, $a = g/3 = 3.27 \text{ m/s}^2$; $T = 13.1 \text{ N}$.
6	The string is light and the pulley is smooth.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.